import streamlit as st import pickle import pandas as pd # Load Model with open("Linear_Regression_california.pkl", "rb") as f: model = pickle.load(f) # Page Config st.set_page_config(page_title="California House Price Predictor", layout="centered") # Title st.title(" California House Price Predictor") st.subheader("Developed by Shrirang") st.write("Enter property details to predict the house value.") # Sidebar Information st.sidebar.header(" Feature Description") st.sidebar.write(""" **Median Income** → Average income of households in the area. **House Age** → Age of houses in the area (years). **Area Population** → Total population in the area. **Average Occupancy** → Average number of people living in a house. **Average Bedrooms** → Average number of bedrooms per house. **Property Latitude** → Geographical latitude of the property location. """) # Input Fields median_income = st.number_input(" Median Income", min_value=0.0, value=5.0, step=0.1) house_age = st.number_input(" House Age (Years)", min_value=0.0, value=20.0, step=1.0) area_population = st.number_input("Area Population", min_value=0.0, value=1000.0, step=100.0) average_occupancy = st.number_input(" Average Occupancy", min_value=0.0, value=3.0, step=0.1) average_bedrooms = st.number_input(" Average Bedrooms", min_value=0.0, value=1.0, step=0.1) property_latitude = st.number_input(" Property Latitude", value=34.0, step=0.01) # Prediction if st.button("Predict House Value"): input_data = pd.DataFrame([[median_income, house_age, area_population, average_occupancy, average_bedrooms, property_latitude]], columns=["MedInc", "HouseAge", "Population", "AveOccup", "AveBedrms", "Latitude"]) prediction = model.predict(input_data) house_price = prediction[0] * 100000 st.success(f" Predicted House Value: ${house_price:,.2f}") st.info("The predicted value is based on the trained California Housing Linear Regression model.")